

To What Extent Does Intrinsic Versus Extrinsic Motivation Predict Long-Term Athletic Performance in Adolescent Athletes?

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Abstract

This paper investigates the extent to which intrinsic versus extrinsic motivation predicts long-term athletic performance in adolescent athletes. Drawing upon Self-Determination Theory (SDT; Deci & Ryan, 1985, 2000), Cognitive Evaluation Theory (CET), and a synthesis of empirical research spanning multiple sports and cultural contexts, this study evaluates how each motivational orientation influences performance outcomes, sport persistence, burnout, and dropout rates.

Data from landmark studies including a prospective 492-player soccer cohort (Calvo et al., 2010), the Sport Motivation Scale-6 validation study (Mallett et al., 2007), a six-year longitudinal cohort (Valenzuela-Moss et al., 2024), and a large-scale Turkish high school sample (PLOS One, 2025) are synthesised and analysed.

Statistical evidence consistently indicates that intrinsic motivation is a significantly stronger positive predictor of long-term athletic performance than extrinsic motivation. External regulation and amotivation predict sport dropout, reduced engagement, and burnout, particularly during the critical transition years of adolescence (ages 13-17). Alarming, 70% of young

athletes drop out of organised sport by age 13, a crisis largely attributable to externally-pressured motivational climates.

The overjustification effect and the self-determination continuum are presented as explanatory frameworks. The paper concludes that, while extrinsic motivation may support short-term performance gains, intrinsic motivation is the superior and more durable predictor of long-term athletic success. Practical implications for coaches, parents, and sport organisations particularly within the Indian adolescent sporting context have also been discussed.

Keywords: *intrinsic motivation, extrinsic motivation, adolescent athletes, long-term performance, Self-Determination Theory, sport dropout, burnout, motivational climate*

1. Introduction

Athletes are motivated to train and compete for a variety of reasons and the type of motivation that an individual has can play a significant role in determining their long-term participation in sport. Motivation that is derived from within the athlete (intrinsic) as opposed to external Pressures (extrinsic) has been a topic of considerable interest to researchers and has been shown to play a critical role in the athletes long-term participation in sport. For the adolescent athlete in particular, participation in sport can play a significant role in their physical, social and emotional development and the reasons that they participate in sport can have a considerable impact on the experience that they have and the level of involvement that they go on to have in the future.

For many young athletes their adolescent years are crucial in the formation of their athletic identity. Typically during adolescence athletes experience rapid physical change, increasing social awareness, and an attempt to establish a coherent sense of self (Harter, 1999). For millions of children and adolescents participating in organized sports around the world, their involvement in sport will either become a lifelong passion or be a phase of life that they grow out of. Interestingly, statistics suggest that the majority of children involved in organized sport quit by the age of 13, with the American Academy of Pediatrics (2024) reporting that 70% of children involved in sport drop out by the age of 13. Such a high level of attrition is not primarily caused by children's physical development but is more likely to be the result of a range of motivational factors.

The main research question that this paper will attempt to answer is: How does intrinsic motivation compare with extrinsic motivation in determining long-term athletic performance in adolescent athletes.

In answering this question the coach, the parents of the athlete and the sport psychologist working with adolescents in competitive sports can have real-world application of their work with young athletes.

A coach could build a squad that would last longer than one made up of athletes who are only motivated by external rewards. Parents would be alerted to the over-justification effect that could be killing their child's love of sport by over-rewarding them for participating in sport. Above all, the sport psychologist could develop training programs for young athletes within a motivational climate that is proven to promote optimal levels of intrinsic motivation. This would be the case whether the sport psychologist is based in cricket academies in Mumbai or football clubs in Chennai.

The rest of the paper is organized as follows. Section 2 introduces the theoretical background to the research question posed in this paper, focusing on Self-Determination Theory and on the sub-theory of Cognitive Evaluation Theory. In Section 3, we turn to an overview of the available literature that informs this paper's primary question. In Section 4 of this paper, the quantitative data provided in this research will be presented. In Section 5 of the paper, the available data will be discussed, implications of the findings highlighted, and limitations of the evidence reviewed. In the final section, conclusions to this research will be outlined, along with recommendations that the available data supports for athletes, their parents, their coaches, and their sport psychologists. The Bibliography that follows this end of the paper is formatted to APA 7th edition standards.

2. Theoretical Framework

2.1 Self-Determination Theory (SDT)

The theoretical underpinning for this study is Self-Determination Theory (SDT). SDT is a macro-theory that explains human behavior and motivation by outlining three innate psychological needs that are present in all individuals. These basic psychological needs are described as autonomy, competence and relatedness. When these basic psychological needs are satisfied, individuals are said to be self-determined and act with autonomy; that is they behave in a manner that is congruent with their true values and that they have endorsed (Deci & Ryan, 2000).

Self-Determination Theory organizes motivation along a continuum, from amotivation to intrinsic motivation. This continuum is made up of a number of types of extrinsic motivation.

Table 2 lists these in order of most to least self-determined (i.e. from most to least motivating as athletes persist, achieve and report positive outcomes as a result of participation in physical activity).

Research into the causes of persistence and achievement in sport has consistently found that athletes who are self-determined (i.e. motivated by reasons that are endogenous to themselves) are more likely to continue participating, achieve their goals, and report positive affect than athletes who are not. Importantly, however, SDT does not portray all forms of extrinsically motivated behavior as being equally likely to lead to undesirable outcomes. In fact, identified and integrated regulation are seen as forms of extrinsic motivation that can be as positive as intrinsic motivation. It is the more controlling forms of extrinsic motivation (i.e. external regulation and, to a lesser extent, introjected regulation) that are thought to have negative effects in the long term.

Table 1

The Self-Determination Continuum Applied to Adolescent Sport (Adapted from Deci & Ryan, 2000; Mallett et al., 2007) Nitya Jimmy Mahtani Nitya Jimmy Mahtani

Regulation Type	Category	Example in Sport	Degree of Self-Determination	Long-Term Performance Effect
Intrinsic	Intrinsic Motivation	Playing cricket for the joy of mastery	Highest	Most Positive
Integrated	Extrinsic (most	Training aligns	Very High	Positive

	autonomous)	with one's identity as an athlete		
Identified	Extrinsic	Practising because fitness is personally important	Moderate-High	Moderately Positive
Introjected	Extrinsic	Training to avoid guilt or shame	Low-Moderate	Neutral to Negative
External	Extrinsic (least autonomous)	Competing only for trophies or parental approval	Low	Negative (long- term)
Amotivation	Non-regulation	No reason to participate; disengaged	None	Highly Negative (dropout risk)

Table 1 illustrates a critical insight: the motivational type that predicts the most positive long-term performance outcomes is intrinsic motivation, followed by the more autonomous forms of extrinsic regulation. The more controlling forms of external regulation—and especially amotivation—consistently predict negative long-term outcomes, including dropout and burnout (Calvo et al., 2010).

2.2 Cognitive Evaluation Theory (CET)

A sub-theory of SDT, Cognitive Evaluation Theory (CET; Deci & Ryan, 1985) addresses how external events affect intrinsic motivation. CET proposes that events which support an individual's sense of autonomy and competence—such as informational feedback and autonomy-supportive coaching—tend to enhance intrinsic motivation. Conversely, controlling events—such as surveillance, contingent rewards, and imposed deadlines—undermine intrinsic motivation by diminishing perceived autonomy.

CET is particularly relevant in the context of youth sport because it explains a counter-intuitive but well-documented phenomenon: the introduction of extrinsic rewards for activities that an individual already finds intrinsically rewarding can actually reduce motivation for those activities. This is the overjustification effect (Lepper et al., 1973; Deci, 1971), discussed in detail in Section 3.3.

2.3 Achievement Goal Theory

An alternative framework for understanding the impact of sport environments is provided by Achievement Goal Theory (AGT). AGT (Ames, 1992; Nicholls, 1989) proposes that sport environments can be characterized as either mastery oriented or performance/ego oriented. The mastery-oriented approach emphasizes personal growth and improvement through effort and learning, and is typically associated with an intrinsic motivational orientation. Research utilizing AGT has found that environments that are more mastery oriented (i.e., promote an intrinsic motivation orientation) encourage persistence, enjoyment of physical activity, and superior long-term performance when compared with compared to performance/ego-oriented environments (Ames, 1992). In contrast, environments that are more performance/ego oriented (i.e., emphasize

external regulation such as winning and comparative performance with others) promote anxiety and decrease persistence in the face of failure. Ultimately, children and adults who participate in such environments are at risk for early sport dropout.

3. Literature Review

3.1 Intrinsic Motivation and Athletic Performance: What the Evidence Shows

Intrinsic motivation for sport is a well established variable to show superior performance in sport. Research demonstrates that individuals who are motivated by intrinsic reasons for participating in physical activity are more likely to be engaged in physical activity than individuals who are motivated by extrinsic reasons (Deci & Ryan, 2000).

Research also has consistently shown that individuals who are motivated by intrinsic reasons for participating in physical activity spend more time involuntarily practicing their sport, concentrating on their physical skills, coping with negative experiences that occur during participation in physical activity, and are likely to be physically active for longer periods of time during individuals' adolescents (Vallerand, 1997).

The evidence for the link between intrinsic motivation and better sport performance is extensive and consistent. The main behavior characteristic of intrinsically motivated athletes is that they engage in the sport for reasons related to the experience itself. They are more likely than other athletes to invest time in skill related tasks, to concentrate more during practice, to react more constructively to adversity, and to remain engaged in the sport for longer periods of time during the adolescents years (Deci & Ryan, 2000; Vallerand, 1997).

As Mallett et al. (2007) validated the Sport Motivation Scale-6 (SMS-6) scores for competitive athletes, they found that the factor for intrinsic motivation of the SMS-6 correlated strongly positive with flow experiences during sport as reported by the Dispositional Flow Scale-2 (DFS-2, $r = +0.61$, $p < .05$). In contrast to intrinsic motivation, scores for extrinsic motivation and amotivation did not correlate or correlated negatively with DFS-2 scores (Mallett et al., 2007).

Further support for this finding comes from a 2025 cross-sectional study that included 1,399 physically active high school students (aged 14–18) from eight different physical education settings in Turkey. In this study, participants completed the Intrinsic Motivation Inventory (IMI) and reported on their achievement emotions following a physical education lesson.

The findings from this study supported the hypothesized model using Structural Equation Modelling (SEM). Intrinsic motivation was found to strongly predict positive achievement emotions ($r = +0.67$, $p < .001$) and to significantly and negatively predict negative achievement emotions. Extrinsic motivation was found to only weakly correlate with positive achievement emotions ($r = +0.21$) and not to significantly correlate with negative emotions.

In conclusion, the findings of this study provided further support for the finding that intrinsic motivation leads to long-lasting meaningful engagement in physical activity, whereas external motivators only lead to short-term gains in performance that are not sustainable in the long term (PLOS One, 2025).

A study from Bryant University conducted a mediation analysis of data from 136 Division I athletes from Roberto et al. (in press) utilizing the Preacher and Hayes (2008) approach. Intrinsic motivation was a significant independent predictor of athletes' athletic

performance as rated by their coaches ($r = +0.48$) and locus of control did not achieve significance as a predictor of athletes' performance, whether or not it was mediated by intrinsic motivation. This finding is important as it indicates that intrinsic motivation is not just a personality trait or characteristic that athletes are born with and that athletes can develop. Intrinsic motivation can also be an independent predictor of high levels of athletic performance.

3.2 Extrinsic Motivation: Short-term Gains, Long-term Costs?

External motivators such as unexpected rewards and incentives can be very powerful at boosting short-term performance when they are seen as providing information related to an athlete's competence and are not controlling. There is empirical support for such findings dating back to Deci et al. (1999). They found that adding tangible external rewards to baseline levels of intrinsic motivation did not decrease intrinsic motivation when such rewards were perceived to be providing information related to an athlete's competence and were not seen as controlling (also referred to as the CET model of autonomy support; G. H. Gallup & J. W. Wilson, 1980; J. W. Wilson, 1978).

External regulation or participation in sport for external rewards (e.g. trophies, prize money, ranking, selection for future events) or to avoid negative stimuli (e.g. coach's criticism, parents' disappointment, social exclusion from team or group) is the most problematic form of extrinsic motivation in adolescent sport. Research has identified that this type of motivation is associated with a range of negative outcomes for athletes in the long-term. For example, Appleton and Hill (2012) found that athletes who were motivated by external rewards were socially prescribed perfectionists, i.e. they were motivated by externally imposed standards to gain social recognition from others. These athletes reported higher levels of burnout as measured by the Athlete Burnout Questionnaire (ABQ).

Gillet, Berjot and Gobance (2009) looked at the motivation of 90 competitive athletes in various sports and found a short term increase in performance that was based on an external motivational model (controlled regulation) which is likely to struggle to sustain performance in the long term. Utilizing the hierarchical model of intrinsic and extrinsic motivation (HMIEM) developed by Vallerand (1997) the researchers found that when athletes were motivated by external rewards for their performance there was a decrease in performance once the external reward was removed.

The study of youth single-sport, multi-sport and non-specialised adolescent athletes (Crouch et al., 2022) found that a focus on extrinsic factors for involvement in and participation in sport can cause athletes to believe that their behavior is being controlled by the external payment of rewards. This in turn negatively affects the athletes' intrinsic enjoyment of the sport. In contrast, non-specialised athletes scored higher on intrinsic motivation than their single- and multi-sport counterparts. This is not to say that non-specialised adolescents are more motivated than their single- and multi-sport counterparts in the short-term; however, it does suggest that the involvement of non-specialised adolescents in sport is more sustainable in the long-term.

3.3 The Overjustification Effect

The overjustification effect is perhaps the most significant phenomenon at the interface of intrinsic and extrinsic motivation. In 1971, Deci first described the overjustification effect when he reported a study in which university students who were paid to solve jigsaw puzzles that they had been working on for intrinsic reasons showed a decline in their intrinsic motivation to work on the same puzzles after payment had ceased. In fact, their level of intrinsic motivation to work on the puzzles decreased below their level of intrinsic motivation prior to the payment. Lepper, Greene and Nisbett (1973) explained the overjustification effect in terms of an attributional

redefinition of behavior. When an individual receives an expected external reward for his or her behavior, he or she attributes his or her behavior to the external reward rather than to his or her own internal motivation. Thus, the external reward decreases the individual's subsequent intrinsic motivation for the same activity.

In Deci's (1971) study he paid university students to complete jigsaw puzzles which they had previously been working on for money. After he had stopped paying them, they showed less interest in completing the puzzles than they had previously when they had been working on them for money in fact they showed less interest than when they were first working on the puzzles for money, i.e. at their baseline level. It appears that by paying the students for working on the puzzles Deci had altered their reasons for completing the puzzles and had, as a result, decreased their intrinsic motivation for completing them.

Further support for the application of the overjustification effect to adolescent sport comes from observation of their natural environment. Over the last two decades there has been an increase in the number of athletes who compete at elite levels of sport at an early age. In addition to representing their clubs and teams at high levels, young athletes are also involved in activities such as trying out for elite squads, entering talent identification programs, seeking sport scholarships, and participating in media coverage of their sport.

All of these activities focus on the end outcomes of sport, which can create a performance-oriented motivational climate. This type of environment can have a negative impact on young athletes who initially entered into sport for reasons related to intrinsic motivation. Once the focus is placed on the DT as the theoretical framework, they found that dropout was statistically explained by higher levels of amotivation ($r = -0.63$, $p < .001$), external regulation ($r = -0.41$, $p < .001$), and introjected regulation, combined with lower satisfaction of autonomy and

relatedness needs. Persistence in sport, by contrast, was predicted by higher levels of intrinsic motivation ($r = +0.54$, $p < .001$) and need satisfaction. These findings are among the most methodologically rigorous in the adolescent sport literature and provide strong causal evidence for the role of intrinsic motivation in long-term athletic persistence.

A six-year longitudinal study by Valenzuela-Moss et al. (2024), following the graduating class of 2023 from 7th to 12th grade, tracked changes in sports participation, specialisation, and burnout using mixed-effects regression models. The results were stark: the number of days per week in which students engaged in at least 60 minutes of strenuous exercise progressively decreased from 4.0 days/week in 7th grade to just 2.3 days/week in 12th grade ($p < 0.01$). Burnout scores increased monotonically across grade levels, with specialisation intensity partially mediating this relationship. The authors identified the transition from middle school to high school as a critical window during which externally-driven performance pressure—school sports selection, parental expectations, college recruitment considerations—begins to overwhelm intrinsically-motivated participation.

Self-determination theory burnout research adds a clinical dimension. Amorose and Anderson-Butcher (2007), cited in Nova Southeastern University research (NSU, 2024), found that autonomy support from coaches was a critical buffer against burnout in high school athletes. When coaches adopted controlling behaviours prescribing training without explanation, using conditional positive regard, emphasising winning over development athlete burnout was significantly elevated. Conversely, coaches who supported athlete autonomy, explained the rationale behind training decisions, and acknowledged athletes' perspectives produced lower burnout and higher persistence rates, consistent with SDT's predictions.

3.4 Gender, Cultural Context, and Motivational Patterns

Why are young athletes motivated to train and compete in their chosen sport? The answer to this question may be different for every young athlete and will be influenced by the athlete's gender, culture and sport. A recent study carried out with teenage bowlers from across Malaysia found that the majority of young athletes are motivated by internal forces that are associated with their love for a particular sport. A total of 240 young athletes (with an average age of 16.61 years) from a variety of different backgrounds took part in the study and the results revealed some very important differences between the boys and girls that took part.

For instance, those boys that compete at the highest levels are more likely to be externally motivated by factors such as receiving praise from others. In contrast the girls that took part in the study were more likely to be motivated by their personal goals and values, thus finding their own internal drive to succeed without the need for external pressure.

Importantly the girls involved in the study scored higher on aspects of motivation which highlights that the driving force behind their participation is to learn and to improve. Therefore, the study highlights that motivation to participate in sport is a complex process that is influenced by a wide range of different factors and that there will be many different reasons that drive each athlete. However, there is evidence to suggest that those who are internally motivated by their love for a sport are more likely to be successful in the long term.

In the Indian context, the motivational landscape of adolescent sport is particularly complex. India's sporting culture is characterised by intense performance expectations in a limited number of high-status sports (cricket, badminton), juxtaposed with institutional under-investment in grassroots sport development. Studies from South Asian contexts suggest that

parental pressure a form of identified extrinsic regulation at best, external regulation at worst plays an outsized role in adolescent athletes' motivational orientation.

For Indian students navigating both academic pressure and sporting aspirations, the balance between intrinsic enjoyment and externally-imposed expectations is a critical determinant of whether sport participation is sustained through adolescence.

With external rewards for performance, young athletes can begin to lose their intrinsic motivation for participation in the activity. Although the external rewards may be removed at some point (i.e., not selected for the elite squad, do not receive a scholarship, do not perform at a level to receive media recognition), the athlete will no longer have any motivation for participation in the activity, resulting in early dropout from sport. The Association for Applied Sport Psychology (AASP) states that when an overemphasis on extrinsic motivation exists, athletes perceive that their behavior is under the control of the external reward, thus being removed from the inherent enjoyment of the sport.

The overjustification effect has also been found to have more negative effects on intrinsic motivation as a result of extrinsic rewards on children as opposed to college students or adults (Tang & Hall, 1995). The work of Deci and his colleagues (Deci et al., 1999) indicates that children lack the ability to make a clear distinction between their intrinsic motivation and the external rewards that have been provided for their behavior. This is particularly relevant when designing reward systems within youth sport.

3.5 Sport Dropout, Burnout, and Motivational Climate

It's really important to understand what motivates young athletes to stay in or drop out of sports. This is a big deal because a lot of kids quit playing sports at a pretty young age. In the US, for example, over 60 million kids play organized sports, but a whopping 70% of them stop

by the time they're 13. That's according to the American Academy of Pediatrics. Similarly, in Australia, a study found that while almost half of kids aged 5-14 play sports, that number drops to less than a quarter by the time they're 15-19.

This is a big problem because sports can be really good for kids, teaching them important skills like teamwork and perseverance. So, figuring out why kids drop out and how to keep them motivated is crucial. By understanding what drives kids to stay in or leave sports, we can work on creating a more supportive and engaging environment that keeps them playing and having fun.

The most comprehensive prospective study of motivational predictors of dropout is Calvo et al. (2010), who followed 492 soccer players aged 13 to 17 over a one-year period. Using S

3.6 Gender, Cultural Context, and Motivational Patterns

Motivational patterns in adolescent athletes are not alike. Research indicates that there are a number of variables that affect adolescent athletes including gender, cultural context, and the type of sport in which they participate. A recent study conducted by Aziz and Justine (2015) focused on 240 adolescent ten-pin bowlers from Malaysia with a mean age of 16.61 years. The total intrinsic and extrinsic motivation scores for the sample of participants revealed significant differences with the athletes' intrinsic motivation scores far exceeding their scores for extrinsic motivation.

In terms of gender differences on specific subscales it was noted that the athletes' scores for intrinsic motivation to know, to accomplish, and to experience stimulation as well as their identified regulation scores were significantly different. Male athletes in competitive contexts scored higher on the external regulation subscale than their female counterparts. In contrast female athletes scored higher on the identified regulation subscale, which suggests that

adolescent female athletes are more likely to identify with aspects of sport that they believe are valuable and are therefore able to motivate themselves to participate in physical activity independently of external pressure. It appears that both genders are more motivated by intrinsic orientations.

The adolescent sport context in the Indian scenario is very complex. While the country is obsessed with performing well in a few high profile sports like cricket and badminton, and has high profile coaches and sports infrastructure, the grass root sports development is grossly under-invested. Most of the research in South Asian context suggests that parental pressure is a major factor that influences adolescent athletes' motivational orientation. The motivational orientation of the adolescent athletes who are also students would thus depend on whether they are able to experience the intrinsic enjoyment of sports or are pressurized by external factors to perform. The fear of this external pressure could potentially threaten the participation of adolescents in sports throughout their life.

4. Data Analysis

4.1 Overview of Data Sources and Synthesis Approach

Quantitative analysis in this section draws upon published data in terms of correlation coefficients, participation data and performance data for a number of time points for a number of key studies reviewed in Section 3. Pearson correlations (r) between motivational variables and performance variables have been synthesized where studies reported these statistics. It should be noted that while the findings reported in this section are based upon reported statistics from individual studies a formal meta-analytic procedure is not undertaken in this paper. However, the

consistency of findings of a specific direction in relation to a particular comparison across a number of different cultures provides strong convergent validity for findings reported in this paper.

4.2 Summary of Key Studies

Table 2

Summary of Key Empirical Studies on Motivation and Adolescent Athletic Performance

Study	Sample	Design	Key Finding	Implication
Calvo et al. (2010)	492 soccer players (13-17 yrs)	Prospective, 1-year follow-up	Amotivation & external regulation predicted dropout	SDT-based coaching reduces attrition
Mallett et al. (2007)	Multi-sport athletes (SMS-6 validation)	Cross-sectional psychometric	IM strongly correlated with flow states; EM negatively/not at all	Intrinsic motivation underpins peak performance states
AAP Report (2024)	60 million US youth	Clinical meta-review	70% drop out by age 13; burnout	Overprofessionalisation undermines self-

Study	Sample	Design	Key Finding	Implication
	athletes		linked to extrinsic pressure	determination
Valenzuela- Moss et al. (2024)	77 students, 7th to 12th grade (6-year longitudinal)	Prospective longitudinal	Exercise days dropped 4.0 to 2.3/week; burnout increased through high school	Transition from middle to high school is a critical attrition window
PLOS One (2025)	1,399 Turkish high school students	Cross- sectional SEM	IM strongly predicted positive achievement emotions; EM weak/non- significant	Motivational climate design should prioritise autonomy support
Bryant Univ. Study (Roberto et al.)	136 Division I athletes	Quantitative, mediation analysis	IM significant predictor of athletic	IM acts as independent predictor beyond personality traits

Study	Sample	Design	Key Finding	Implication
			performance; locus of control not significant alone	

Table 2 synthesises the six most methodologically significant studies reviewed. A consistent pattern emerges across studies employing cross-sectional, prospective, and longitudinal designs stating that intrinsic motivation is positively and significantly associated with beneficial athletic outcomes (performance, persistence, positive emotions, flow states), while external regulation is consistently negatively associated with these outcomes.

4.3 Correlation Data: Motivational Orientation and Performance Outcomes

Table 3 presents synthesised correlation coefficients between three motivational orientations—intrinsic motivation, external regulation, and amotivation—and five key performance-related outcome variables, drawn from the primary studies reviewed.

Table 3

Synthesised Correlation Coefficients: Motivational Orientation and Athletic Outcomes ($p < .05$; ns = non-significant)*

Outcome Variable	Intrinsic Motivation (Correlation r)	External Regulation (Correlation r)	Amotivation (Correlation r)	Source
Sport Persistence (1-year)	+0.54*	-0.41*	-0.63*	Calvo et al. (2010)
Flow State (DFS-2)	+0.61*	-0.12 (ns)	-0.28*	Mallett et al. (2007)
Positive Achievement Emotions	+0.67*	+0.21*	-0.55*	PLOS One (2025)
Athletic Performance (Coach-rated)	+0.48*	+0.19 (ns)	-0.44*	Roberto et al. (Bryant Univ.)
Sports Engagement (Adolescents)	+0.59*	+0.23*	-0.51*	SMS-II Validation (2013)

Table 3 reveals a rather clear pattern of association, intrinsic motivation was found to be positively related to all five indicators at moderate to strong levels of association ($r = +.48$ to $+.67$). The average positive correlation of intrinsic motivation with the various indicators was about $r = +.58$ ($p < .05$).

The correlations of external regulation with the indicators showed weak to moderate positive association with the indicators of engagement and positive emotions, but were not

significant for the indicators of coach rated performance, flow and sport persistence in most of the studies, i.e. in contrast to intrinsic motivation, external regulation does not seem to predict the indicators of performance in sports in the long run.

These indicators are influenced in the short run, but in a positive way, by external regulation. Amotivation was found to be negatively related to all indicators of motivation at moderate to strong levels of association ($r = -.28$ to $-.63$). The associations of amotivation with the indicators of persistence in sport and of positive achievement emotions were found to be the most negative ones.

The difference in predictive power between intrinsic motivation and external regulation is most dramatic for sport persistence and the flow states that the two variables are most predictive of actual long-term athletic development. This suggests that intrinsic motivation does not merely correlate with athletes feeling happier, it fundamentally structures the psychological states that enable peak performance and sustained engagement over time.

4.4 Sport Dropout Statistics: The Scale of the Motivational Crisis

Table 4

Youth Sport Dropout and Participation Data Across Age Groups

Age Group / Grade	Sport Participation Rate	Dropout / Burnout Indicator	Source
Ages 5-14	45-46%	Peak participation	Australian participation data (Calvo et al., 2010)

Age Group / Grade	Sport Participation Rate	Dropout / Burnout Indicator	Source
Age 13 (US)	~30% (retained)	70% dropout rate by age 13	AAP (2024); Changing the Game (O'Sullivan)
Ages 15-19	<50% (Europe)	Steady decline below threshold	European secondary education data (PMC, 2024)
Grade 7 to Grade 12 (US)	Declined 4.0 to 2.3 exercise days/week	Burnout scores increased longitudinally	Valenzuela-Moss et al. (2024), Orthopaedic Journal

Table 4 outlines the demographics surrounding the adolescent sport dropout crisis. The trends highlighted in the table indicate a steep decline in participation in sport by adolescents. The steepest decline in participation occurs between the ages of 13 and 17 years of age. It is during this critical developmental period that the motivational climates created by coaches and the sporting environment become increasingly performance-oriented and driven by external outcomes (Valenzuela-Moss et al., 2024).

The findings from a six-year longitudinal study of adolescents participating in organized sport highlight trends of decline in the number of days per week that adolescents engage in exercise. Furthermore, the findings highlight a trend of increase in adolescent burnout over the six years of secondary education.

The trend of early dropout from sport by adolescents is highlighted by the American Academy of Pediatrics (2024) who reported that 70% of children who play sports drop out by the age of 13 years. This is not an isolated finding and supports data from other countries including Australia (Calvo et al., 2010) and Europe (PMC, 2024). The current structure of youth sport is not motivated or sustainable for long-term participation in physical activity.

The 70% dropout rate by age 13 found by the American Academy of Pediatrics (2024) is not an isolated instance, similar findings have found high dropout rates of children playing organized sport among young people in Australia (Calvo et al., 2010) and those from across Europe (PMC, 2024). A systematic change in practice within youth sport is warranted given that 70% of children who start sport at the age of 5 are dropout of organized sport by the time that they are 13 years of age. This apparent epidemic of ‘dropout’ of children playing organized sport is attributed to the motivational unsustainable effects of a sporting environment that facilitates early sport specialization and is primarily driven by a performance oriented, outcome focused coaching climate.

4.5 Interpreting the Data: Intrinsic Motivation as the Superior Predictor

Taken together, the data presented in Tables 1-4 support a clear conclusion: intrinsic motivation is a significantly stronger predictor of long-term athletic performance than extrinsic motivation in adolescent athletes. This conclusion rests on four converging lines of evidence.

First, correlation analyses consistently show moderate-to-strong positive relationships between intrinsic motivation and all key performance outcome variables, while external regulation shows weak or non-significant relationships with the most critical outcomes (flow, persistence, coach-rated performance).

Second, longitudinal data (Calvo et al., 2010; Valenzuela-Moss et al., 2024) demonstrate that intrinsic motivation predicts sustained sport participation over 12-month and 6-year follow-up periods, respectively, while external regulation and amotivation predict dropout.

Third, burnout and dropout statistics demonstrate that the dominant mode of youth sport characterised by early specialisation, outcome-focused coaching, and extrinsic reward emphasis produces attrition rates of up to 70% before adolescence even properly begins. This catastrophic dropout rate is consistent with the predictions of SDT and CET that externally-controlled motivational environments undermine the self-determination that sustains long-term engagement.

What happens when we use rewards to motivate people, especially kids and teenagers, is that it can actually have the opposite effect. Research has shown that giving external rewards can undermine people's natural desire to do something. This is known as the overjustification effect. Studies have proven that rewards don't just fail to support internal motivation, but they can also damage it. This is especially true for young people. So, when we use rewards to motivate kids and teenagers in sports, it's not just a neutral thing - it can actually be harmful in the long run.

5. Discussion

5.1 Implications for Coaching Practice

The evidence reviewed in this paper has direct and urgent implications for coaching practice at the adolescent level. The single most impactful change a coach can make to improve the long-term athletic development of their athletes is to shift from a controlling, outcome-focused motivational climate to an autonomy-supportive, mastery-oriented one. In concrete

terms, this means providing athletes with meaningful choices in their training, explaining the rationale behind drills and sessions, offering competence-affirming feedback that emphasises personal progress rather than comparative ranking, and creating a psychologically safe environment in which effort and improvement are valued above results (Amorose & Anderson-Butcher, 2007).

The reward structures coaches use must be designed with awareness of the overjustification effect. Unexpected, informational rewards praise that communicates specific competence enhance or preserve intrinsic motivation. Expected, contingent, controlling rewards prizes tied to winning outcomes, public recognition tied to rankings risk displacing intrinsic motivation with external regulation. This distinction matters because once extrinsic motivation becomes the primary driver, the coach has created a structurally fragile athlete: one whose engagement depends on the continued provision of external rewards and who may be devastated motivationally and emotionally when those rewards are removed.

5.2 Implications for Parents

Research supports the claim that parents are among the most powerful determinants of an adolescent athlete's motivational orientation. When parents allow their children to develop in their own way, they foster an atmosphere where children can find intrinsic motivation for involvement in physical activity. In contrast, controlling parental behavior that places pressure on children to participate in or perform in sports is often associated with introjected and external motivation (both of which have been linked to burnout and early sport termination). Children who receive unconditional positive regard from their parents are more likely to be intrinsically motivated and to persist for longer periods of time in physical activity. Furthermore, parents who

are interested in their children's experiences and allow their children to have a say in sports-related decisions are also more likely to foster intrinsic motivation in their children.

The findings of the research have huge implications for India, where secondary school (Classes 9-12) is a time of great academic pressure. A lot of parents make decisions about where to send their children to college, based on resources that could also be used to develop their child's sporting ability. If parents view sport solely in terms of extrinsic rewards (e.g. college scholarships, prestige of job, parental status), then the child's intrinsic motivation to participate in sport is likely to decrease, and lead to burnout and / or early sports dropout. On the other hand, if parents view sport in terms of growth, fun, health and teamwork then their child is likely to continue participating in sport throughout adolescence and into adulthood.

5.3 Implications for Sport Organisations in India

India's sporting infrastructure is at a critical developmental juncture. The success of Indian athletes at Olympic and World Championship levels in recent years in badminton, boxing, wrestling, and shooting has driven increased investment in high-performance sport. However, the pipeline of adolescent athletes that feeds this high-performance system requires careful attention to motivational ecology. Talent identification programmes that are exclusively outcome-focused risk selecting for extrinsically-motivated athletes while overlooking intrinsically-motivated athletes whose performance ceiling may be higher but whose development is less immediately visible.

The evidence from this paper recommends that Indian sport organisations and academies invest in autonomy-supportive coaching education, design training environments that prioritise mastery climates over performance climates, and delay specialisation and intense selection pressure until the late adolescent years (16+), when athletes have had the opportunity to develop

robust intrinsic motivation foundations. The Youth Sports Trust and academic sporting bodies in India would benefit from systematic measurement of motivational climate using validated instruments such as the SMS-6 or the Perceived Motivational Climate in Sport Questionnaire (PMCSQ), enabling evidence-based programme design.

5.4 Limitations of the Evidence

Several limitations qualify the conclusions of this review. First, the majority of large-scale empirical studies on this topic have been conducted in Western contexts (United States, Europe, Australia), with limited representation of South Asian adolescent athlete populations. Motivational dynamics may differ across cultural contexts particularly with respect to the role of collectivist values, family honour, and academic-sporting identity conflicts that are more salient in Indian contexts.

It's not always easy to tell which way the relationship between intrinsic motivation and performance goes. Some studies have shown a connection between the two, but it's hard to say for sure which one causes the other. When we look at data from one point in time, it's possible that good performance is actually what's driving intrinsic motivation, rather than the other way around. However, when we follow people over time, we can get a better sense of how these two things are related. A few studies have done this and found some interesting results. For example, one study found that intrinsic motivation can lead to better performance, but it's also possible that doing well can make us more motivated. So, it's likely that the relationship between intrinsic motivation and performance goes both ways.

A key methodological challenge in studies of athletic performance is the fact that different measures used in studies are not necessarily congruent. While some studies focus on the perspectives of coaches and athletes, others assess physical fitness or sport-specific performance

measures. Consequently, between-study comparisons are hampered by the various constructs that are investigated.

To this end, there is a need to understand short-term versus long-term athletic performance and few studies have longitudinally followed groups of athletes over extended periods. Typically, studies follow athletes for 1–2 years and as such are unable to provide a complete overview of the athlete's development. One study did longitudinally follow a group of secondary school aged athletes over a 6 year period (until the end of secondary school) (Valenzuela-Moss & colleagues, 2024). What is required is more research that follows athletes from late adolescence to early adulthood in order to gain an understanding of how they continue to develop as athletes. This is currently a lacking area of research within the field of athletic development.

6. Conclusions and Recommendations

6.1 Conclusions

This study looked at how motivation affects athletic performance in teenagers. It found that motivation that comes from within is a better predictor of long-term success in sports than motivation that comes from outside factors. The evidence from several types of research, including long-term studies, statistical models, and reviews of previous research, all points to the same conclusion.

First, intrinsic motivation is the superior predictor of long-term athletic performance in adolescent athletes. It is positively associated with sport persistence, flow states, positive achievement emotions, and coach-rated performance at moderate-to-strong effect sizes ($r =$

+0.48 to +0.67). These associations hold across sports, genders, and cultural contexts, providing robust convergent validity for the conclusion.

Second, extrinsic motivation specifically external regulation and amotivation—predicts sport dropout, burnout, and reduced engagement, particularly during the critical transition years of adolescence. The 70% dropout rate observed in U.S. youth sport by age 13 is not primarily a physical phenomenon; it is a motivational one, driven by the proliferation of externally-controlled motivational climates in youth sport.

When kids are rewarded for doing something, it can actually make them less interested in doing it for fun. This is called the overjustification effect. It happens when rewards are given for specific results, and it's especially true for young people. So, using rewards to motivate kids in sports can have long-term negative effects, rather than just having no impact.

Fourth, the relationship between motivation and performance is mediated by psychological needs satisfaction, as SDT predicts. Coaching and parenting environments that support autonomy, build competence perceptions, and foster relatedness are the most reliable generators of intrinsically motivated, long-term athletic development.

When we look at what drives young athletes to perform well in the long run, we find that how much they want to play for themselves, rather than for outside rewards, makes a big difference. In fact, wanting to play for personal reasons is a strong predictor of how well they'll do over time. On the other hand, playing mainly for external reasons, like pressure from others, can actually lead to them quitting or burning out. This suggests that intrinsic motivation, or playing for the love of the game, is key to lasting success in adolescent athletes.

6.2 Recommendations

These are some recommendations that I would like to share:

1. Coaches of adolescent athletes should adopt a supportive coaching style and, where possible, train using SDT informed practice as part of coaching education programs. They should promote an environment that helps individuals master the sport, utilizing information feedback and allowing athlete input into training decisions.
2. Recognizing and rewarding talented young athletes in Sport within and across organizations for their efforts and achievements, within and outside of competition, within and across different levels of development and participation.
3. Facts should be shared that help support parents who are motivated to get their children to be physically active and playing sports. Parents actions or lack thereof can have a significant impact on their child's involvement in sports for life. Positive reinforcement will go a long way in supporting a student's interests and passion.

Help adolescents fall in love with physical activities. This can be achieved by not forcing them into professional sports at a very young age, by not allowing them to specialize in only one sport too early, and by enabling them to develop sport specific skills while having fun participating in physical activities. By helping adolescents fall in love with physical activities they are likely to participate in life long physical activities, be physically active on a regular basis, and adopt a physically active lifestyle that will enable them to attain optimal levels of physical fitness and sports performance and thereby attain all physical activity related benefits.

Help the young athletes to fall in love with sports. Let them play, and develop the sports that they like. Help them to become good athletes. Winning a few trophies is of little importance

in the early years of participation in sports. The trophies will come anyway in the long run. What is more important is that the young athletes have fun in the sports they play, are to develop in a healthy way, and become good athletes for life, performing at their best in sports for years to come.

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